Quantum encryption

https://doi.org/10.1038/s41567-026-03169-8

Classically impossible cryptography

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Quantum states cannot be copied, which could enable encryption schemes that are impossible classically. Now, substantial progress has been made towards a practical uncloneable encryption protocol using ideas from quantum information theory.

A key difference between classical and quantum information is the no-cloning principle of quantum mechanics[1,2]. This is a major obstacle for many tasks, such as performing quantum error correction. However, the inability to clone quantum states might also be an asset, by enabling so-called uncloneable cryptography that can achieve classically impossible cryptographic operations. For example, a protocol that encodes messages into quantum ciphertexts that are uncloneable could allow for provably deleting cryptographic data from the internet as well as efficiently detecting whether a data breach has occurred or not. Designing such uncloneable encryption schemes has been one of the most notorious problems in quantum cryptography. Now, writing in *Nature Physics*, Archishna Bhattacharyya and Eric Culf have made substantial progress in designing an uncloneable encryption scheme[3].

Prior to Bhattacharyya and Culf's work, there had been a number of attempts to construct uncloneable encryption. The original scheme[4] designed an uncloneable encryption scheme where the security gap — which is the gap between the success probability achieved by the optimal adversary and the success probability achieved by trivial attacks — is 0.35.

Subsequent work has focused on reducing the security gap further to be negligibly close to zero but often with significant restrictions. For example, one scheme with negligible gap[5] assumed the existence of an idealized cryptographic hash function. Later work[6] removed the necessity of this hash function but required the use of fragile quantum states as decryption keys. A recent scheme without such restrictions was only able to achieve a gap of 0.098 (ref. 7). Bhattacharyya and Culf's result has achieved uncloneable encryption without restrictions, with inverse polynomial security gap.

Security proofs for uncloneable encryption schemes typically involve the theory of games that exploit the 'monogamy of entanglement'[8]. These games formalize the intuition that three parties cannot be simultaneously maximally entangled with each other. There are three parties, Alice, Bob and Charlie, who each have part of a shared quantum state. Before the game begins, a set of different possible measurement bases is chosen. Alice's state is measured with respect to one of these measurement bases. After the measurement has been performed, Bob and Charlie are told which measurement basis was used and are then expected to guess the outcome of Alice's measurement in order to succeed in the game. The value of a monogamy of entanglement game is defined as the success probability of an optimal strategy employed by Alice, Bob and Charlie.

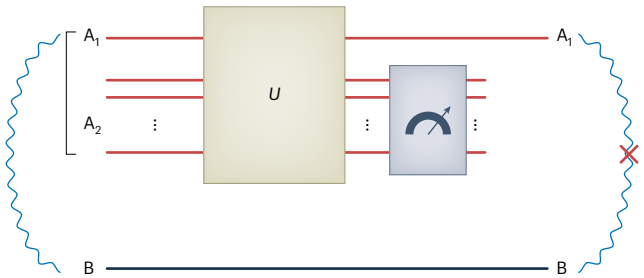


Fig. 1 | An example of entanglement decoupling. Two parties start by sharing a maximally entangled state (indicated by wavy lines). The state of the first party is on registers A_1 and A_2, while the second party's register is denoted by B. The two registers A_1 and A_2 are collectively subjected to a random unitary operation U, mixing quantum information and correlations among the ensemble. The register A_2 is then subjected to a projective measurement. Depending on the strength of quantum correlations between the registers on the original state (prior to applying U), the measurement process destroys the quantum correlations and the entanglement between A_1 and B is effectively eliminated — that is, they have been 'decoupled'.

These games have been useful for modularizing many proofs pertaining to the security of tasks such as quantum key distribution and position verification. Bhattacharyya and Culf proposed and analysed a special class of so-called Haar measure monogamy of entanglement games in which the measurement basis is sampled from the Haar distribution on a carefully chosen set of projection operators. This choice of distribution over projection operators renders it difficult for Alice, Bob, and Charlie to devise a strategy that achieves a high success probability. Using standard teleportation-based techniques, they showed that to demonstrate the existence of an uncloneable encryption scheme, it sufficed to upper-bound the value of the Haar monogamy of entanglement game.

Bhattacharyya and Culf's key insight in proving the upper bound was inspired by the quantum information theory concept of decoupling[9]. When a quantum state that is entangled with an external register is subject to a random unitary quantum operation and then passed through a quantum channel, it becomes decoupled from the external register in a way that depends on the strength of quantum correlation in the entire system (Fig. 1).

In the game considered by Bhattacharyya and Culf, the monogamy of entanglement means that Alice should be far from being maximally entangled with at least one of the parties, say Bob. Then, the decoupling theorem argues that applying a random unitary operation on Alice's register and averaging over some qubits essentially decouples the residual state from Bob's register. Once Bob's register is decoupled, it is uncorrelated with Alice's register and, hence, it is easy to argue that the probability that Bob is able to guess the measurement outcome of Alice's register is low.

In terms of limitations, Bhattacharyya and Culf's work only achieves an inverse polynomial security gap. The ideal goal is to reduce the security gap to be negligible, and this problem is still

open. Another potential disadvantage is that the use of Haar random unitary operations makes their protocol more involved than some of the previous schemes[4] which are simpler and only require quantum algorithms with one round of quantum operations. The proposed scheme is also not noise resilient and, in particular, there are no guarantees on the correctness if some of the qubits are corrupted. This means the scheme cannot be implemented on near-term devices.

Nevertheless, Bhattacharyya and Culf's work moves us closer to the ideal goal of achieving uncloneable encryption with negligible security gap. This could help us make progress in the broader area of uncloneable cryptography. Uncloneable encryption has recently[10] been used in the theoretical construction of copy-protection software, which is another important quantum cryptographic task. Furthermore, the use of the decoupling theorem could potentially inspire other applications in quantum cryptography, especially those applications that use Haar randomness. Whereas Shor's algorithm makes quantum technologies a threat to many legacy encryption protocols that rely on the difficulty of factoring numbers, advances in uncloneable cryptography could help establish a world with information security beyond what is classically possible.

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Published online: 4 February 2026

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Competing interests

The author declares no competing interests.